



VPA: Computer program for the computation of the phase shift in atom–atom potential scattering using the Variable Phase Approach^{☆,☆☆}



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ABSTRACT

A computer code for the computation of the phase shift in atom–atom and electron–atom potential scattering is presented. The phase shift is the central quantity required for the calculation of all types of scattering cross sections. The program uses the Variable Phase Approach (VPA). This is the only exact method for the **direct** calculation of the scattering phase shift. All other methods are based on examining the large distance behavior of the exact solution of the Schrödinger equation. Such methods yield the phase shift only modulo π . The absolute value of the phase shift and its variation with scattering energy is, however, needed for a full understanding of the scattering process, such as for instance in the study of shape resonances and Glory oscillations. The VPA has been sparingly used owing to the instability of the underlying equations and the consequent difficulty of writing computer code to solve them. We present a computer code for the efficient implementation of the VPA method for atom–atom scattering problems over a wide range of scattering energies. The code works for potentials which are singular and for those that are non-singular at the origin. An example of the implementation of the code is given for both an interaction potential with an attractive well and for a purely repulsive potential.

Program summary

Program title: VPA:Variable Phase Approach

CPC Library link to program files: <https://dx.doi.org/10.17632/zh248d2362.1>

Code Ocean capsule: <https://dx.doi.org/10.24433/CO.8146850.v1>

Licensing provisions: GNU General Public License 3 (GPL)

Programming language: Fortran 90

Nature of problem: To compute the quantum mechanical phase shift in an atom–atom or an electron–atom potential scattering problem. The interaction potential, dependent on a single radial coordinate, is defined by the user.

Solution method: The Variable Phase Approach is used. This is the only available method for the direct calculation of the phase shift. The initial internuclear separation for the integration is chosen with the help of the analytical solution for a hard sphere core for all orbital angular momentum quantum numbers L of interest. The first-order nonlinear equation is integrated numerically with a modified LSODA^{19,20} program. The analytic formula for calculation of the Jacobian of the equation is also provided.

Additional comments including restrictions and unusual features: Computer codes for the atom–atom interaction potential, POTENT1, and for its derivative with respect to the inter nuclear separation, DPOTENT1, must be supplied by a user. The calculation of the spherical Bessel and Neumann functions required is performed explicitly for large l angular momentum quantum numbers where the standard upward iterative generation of the functions becomes unstable.

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^{☆☆} This paper and its associated computer program are available via the Computer Physics Communication homepage on ScienceDirect (<http://www.sciencedirect.com/science/journal/00104655>).

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1. Introduction

The Variable Phase Approach relies on the solution of an equation for the phase shift, the so-called phase equation. The phase equation was first published in 1949 [1], then independently obtained in the early 1950s [2–4], and finally analyzed in depth

in 1967 [5,6]. The equation is a first-order nonlinear equation for the scattering phase shift in an atom–atom or electron–atom scattering process. Other methods for the calculation of the phase shift are indirect and obtain it by examining the large separation asymptotic form of the exact solution of the scattering problem. Such methods cannot yield the absolute value of the phase shift but give its value only “modulo π ”. The phase equation has been widely used in the literature for the analysis of non-singular potentials $r^2V(r)_{r \rightarrow 0} \rightarrow 0$, for example, when describing electron–atom interactions [7,8]. However, the equation has not been widely used [9] in scattering theory for singular $r^2V(r)_{r \rightarrow 0} \rightarrow \infty$ potentials, in contrast to the Schrödinger equation, due to the difficulties of numerical implementation, mainly related to the correct choice of the starting point and the accumulation of integration errors.

F. Calogero wrote in his book [5]: “However, although the phase approach is very convenient for the study of the phase shifts produced by a singular potential, it is by no means a good method for their numerical evaluation: the phase equation is very unstable with respect to computational errors because, in the r.h.s, we have almost complete cancellation two large quantities. Thus, any attempt to obtain the scattering phase shifts produced by a strongly singular potential through numerical integration of the phase equation from the origin onward would be very foolish”.

In some situations, the phase shift may jump suddenly over an extremely short distance by a magnitude of π . This occurs particularly for so-called shape resonances and makes the numerical solution of the phase equation very difficult. In this article, we present a computer code that performs the numerical integration of the phase equation for both singular and non-singular potentials with great efficiency.

1.1. General theory: Schrödinger equation

The traditional approach [10,11] for solving the non-relativistic quantum mechanical problem of elastic scattering of one particle by another can be performed by numerical integration of the radial Schrödinger equation for a particle of reduced mass μ_{AB} and a known interaction potential $V(r)$:

$$\frac{d\lambda_l(r)}{dr} + \left[-\frac{l(l+1)}{r^2} + k^2 \right] \lambda_l(r) = U(r) \lambda_l(r) \quad (1)$$

here k is the wave vector, $U(r) = 2\mu_{AB}V(r)$ where $V(r)$ is an interaction potential, l is the orbital angular momentum quantum number and atomic units, in which $\hbar = 1$, have been used. Assuming that the potential has a limited range, and knowing that the homogeneous equation:

$$\frac{d\psi_l(r)}{dr} + \left[-\frac{l(l+1)}{r^2} + k^2 \right] \psi_l(r) = 0 \quad (2)$$

has regular and irregular solutions in the form of Ricatti–Bessel functions of the first and second kinds [12–14], respectively:

$$\hat{j}_l(kr) = kr \times j_l(kr)_{r \rightarrow \infty} \rightarrow \sin\left(kr - \frac{l\pi}{2}\right) \quad (3)$$

$$\hat{n}_l(kr) = kr \times n_l(kr)_{r \rightarrow \infty} \rightarrow -\cos\left(kr - \frac{l\pi}{2}\right) \quad (4)$$

it can be shown that the asymptotic expression for the radial part of the wave function Eq. (1) can be written as:

$$\lambda_l(r)_{r \rightarrow \infty} \rightarrow A_l \sin\left(kr - \frac{l\pi}{2} + \eta_l\right) \quad (5)$$

where A_l is the amplitude of the wave function and η_l is the phase shift for orbital angular quantum number l . It can be obtained by

comparing the numerical solution of Eq. (1) for large radii with Eq. (5).

Having calculated the scattering phase shifts, several different types of cross section may be evaluated. The differential cross section for non-identical particles A and B is given by:

$$\frac{d\sigma}{d\theta}(E, \theta) = \frac{2\pi}{4k^2} \left| \sum_{l=0}^{\infty} (2l+1) (e^{2i\eta_l(E)} - 1) P_l(\cos \theta) \right|^2 \quad (6)$$

where $P_l(\cos \theta)$ is the l th Legendre polynomial and $k = \{2\mu_{AB}E/\hbar^2\}^{1/2}$ is the wave vector. The integral cross section (ICS) is obtained by the integration of Eq. (6) over all polar and azimuthal angles [9]. Other cross sections such as the transport (diffusion) cross section [15] and the viscosity cross section [14] may also be expressed in terms of the scattering phase shift.

1.2. Variable phase approach

The basic idea of the Variable Phase Approach is to change from the Schrödinger equation for a wave function to the equation for the phase shift itself. The simplest way to do it is to substitute the expression:

$$\lambda_l(r) = C_1(r) \hat{j}_l(kr) + C_2(r) \hat{n}_l(kr) \quad (7)$$

into Eq. (1) and then to use the variable constant approach:

$$\hat{j}_l(kr) \frac{dC_1(r)}{dr} + \hat{n}_l(kr) \frac{dC_2(r)}{dr} = 0 \quad (8)$$

And making use of the Wronskian connecting the Ricatti–Bessel functions:

$$W = \left[\hat{j}_l(kr) \frac{d\hat{n}_l(kr)}{dr} - \hat{n}_l(kr) \frac{d\hat{j}_l(kr)}{dr} \right] = k \quad (9)$$

one obtains the general variable phase shift equation (see Ref. [9], Appendix B):

$$\frac{d\eta_l(r)}{dr} = -\frac{1}{k} U(r) F_l^2(\eta_l, kr) \quad (10)$$

$$F_l(\eta_l, kr) = \cos(\eta_l(r)) \hat{j}_l(kr) - \sin(\eta_l(r)) \hat{n}_l(kr) \quad (11)$$

This is a non-linear equation for the phase shift $\eta_l(r)$ which can be solved with either a stiff or a non-stiff solver depending on the nature of potential.

Note that using Eqs. (3)–(5) Eq. (7) may be rewritten as:

$$\lambda_l(r) = A_l(r) \left\{ \cos \eta_l(r) \hat{j}_l(kr) - \sin \eta_l(r) \hat{n}_l(kr) \right\} \quad (12)$$

Thus linking the wave function and the phase shift more directly to each other.

The first two Ricatti–Bessel functions of 0th and 1st orders are given by the simple analytical equations:

$$\begin{aligned} \hat{j}_0(kr) &= \sin(kr) & \hat{j}_1(kr) &= \frac{1}{kr} \sin(kr) - \cos(kr) \\ \hat{n}_0(kr) &= -\cos(kr) & \hat{n}_1(kr) &= -\frac{1}{kr} \cos(kr) - \sin(kr) \end{aligned} \quad (13)$$

and higher orders may be obtained using the two recurrence formulas:

$$\hat{j}_{l+1}(kr) = \frac{2l+1}{kr} \hat{j}_l(kr) - \hat{j}_{l-1}(kr) \quad (14)$$

$$\hat{n}_{l+1}(kr) = \frac{2l+1}{kr} \hat{n}_l(kr) - \hat{n}_{l-1}(kr) \quad (15)$$

Let us now obtain the analytical form of $F_l(\eta_m, kr)$ by substituting Eqs. (13) through (15) into Eq. (11):

$$F_0(\eta_0, kr) = \sin(kr + \eta_0(r)) \quad (16)$$

$$F_1(\eta_1, kr) = \frac{1}{kr} \sin(kr + \eta_1(r)) - \cos(kr + \eta_1(r)) \quad (17)$$

Generalizing this finding we obtain a powerful new recursion relationship:

$$F_{l+1}(\eta_{l+1}, kr) = \frac{2l+1}{kr} F_l(\eta_{l+1}, kr) - F_{l-1}(\eta_{l+1}, kr) \quad (18)$$

Thus we get an approach to the calculation of Eq. (10) twice as fast as before because one needs to compute only one recurrence formula, Eq. (18), instead of two, Eqs. (14) and (15). To the best of our knowledge this is the first time that this proposed approach has been suggested.

There is, however, one important limitation on using Eq. (18). It is well known [16,17] that the direct forward recurrent calculations of spherical Bessel functions are numerically stable only for l less than kr . The function $F_l(\eta_m, kr)$ inherits this property from its antecedent. We therefore use the recurrence relationship of Eq. (18) only for l less than kr and for greater values of l we calculate the spherical Bessel and Neumann functions needed in Eq. (11) directly. The degree of benefit to the computational efficiency from this combination of two methods for the calculations of Eq. (11) provides an advantage which depends on the nature of the interaction potential and relative kinetic energy of the colliding particles. It becomes more beneficial at higher energies when kr is large and a lot of orbital quantum numbers do not exceed its value. The subroutine *F1* implements this mixed approach to the calculation of Eqs. (10) and (11).

1.3. VPA boundary condition

Eqs. (10) and (11) must be solved with a boundary condition of $\eta_l = 0$ at $r = 0$. At $r = 0$ the spherical Bessel function tends to zero while the spherical Neumann functions and possibly the atomic interaction potential go to infinity. Consequently, one needs to choose a radius greater than zero at which the phase shift is negligibly small. It is common to cease calculating phase shifts with increasing values of the orbital angular momentum quantum number at values of 10^{-3} to 10^{-4} when convergence of cross sections is deemed to have been reached. Thus, if we choose the starting value as $\eta_l = 10^{-10}$ our calculations should be safe enough.

The next step is to find a radius r_o at which $\eta_l = 10^{-10}$ to start the calculation. If we assume that at r_o the interaction potential is effectively infinite, then the value of the phase shift at r_o can be obtained analytically using the formula [18]:

$$\tan(\eta_l) = \frac{\hat{j}_l(kr)}{\hat{n}_l(kr)} \quad (19)$$

or, setting the value of $\eta_l = 10^{-10}$ one can approximate the tan function and use:

$$10^{-10} = \frac{\hat{j}_l(kr_o)}{\hat{n}_l(kr_o)} \quad (20)$$

to extract r_o which is different for different k and l values. At $l = 0$ we have the exact solution $\eta_o = 10^{-10} = kr_o$. For $l > 0$ we will apply the approximate solution valid at $kr_o \rightarrow 0$ [6]:

$$10^{-10} \approx \frac{(kr_o)^{2l+1}}{(2l+1)!!(2l-1)!!} \quad (21)$$

where $(2l+1)!! = 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2l+1)$. At $l = 1$ Eq. (21) can be used to find r_o by $kr_o = (3 \cdot 10^{-10})^{1/3}$. Unfortunately, for the larger l values Eq. (21) gives only an upper limit $kr_{\max}$ for kr_o . That is why we applied a binary search of kr_o for an arbitrary l between $kr_{\min}$ and $kr_{\max}$ where $kr_{\min}$ is taken as kr_o for $l-1$. The subroutine *InitPoint* that computes an array of kr_o values for the various l values required is described in the next chapter.

1.4. Jacobian

The computer code for solving a stiff ordinary differential equation requires the Jacobian of the equation. In our case, i.e. that of a first order differential equation the Jacobian is one dimensional and can be easily obtained analytically from Eqs. (10) and (11):

$$J1 = \frac{d(d\eta_l(kr)/dr)}{d\eta_l} = -\frac{2}{k} U(r) F_l(\eta_l, kr) D_l(\eta_l, kr) \quad (22)$$

$$D_l(\eta_l, kr) = -\sin(\eta_l(r)) \hat{j}_l(kr) - \cos(\eta_l(r)) \hat{n}_l(kr) \quad (23)$$

This Jacobian is evaluated by the subroutine named *Jac1*, in our program.

2. Program description

The main task of the code is the numerical integration of the phase equation, Eqs. (10) and (11). We do this using the LSODA [19,20] package a publicly available computer code. The code is able to solve both stiff and non-stiff ordinary differential equations and can switch automatically between them depending on the type of equation under study. The only changes we have made to the LSODA code is that we have increased the number of input variables for it and its inner subroutines by three, namely, the switch of the potential kind KP, the orbital quantum number l and the wave vector k required to calculate Eqs. (10), (11) and (21).

In addition, we have used Barnett's subroutine RICBESS [21] to calculate spherical Bessel and Neumann functions separately.

2.1. Potential subroutine

A subroutine to calculate the interaction potential must be supplied. We have supplied two subroutines as part of the code distribution.

Our main illustrative example uses the Lennard-Jones potential for a collision between two Ne atoms:

$$V(r) = \varepsilon \left(\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right) \quad (24)$$

with parameters $\varepsilon = 29.52 \text{ cm}^{-1}$, $\sigma = 2.756 \text{ \AA}$.

In order to validate the program for a repulsive potential we also include results from an arbitrary repulsive potential:

$$V(r) = \varepsilon \left(\frac{\sigma}{r} \right)^6 \quad (25)$$

where the ε and σ parameters were given the same values as above.

The main program contains a parameter KP in a DATA statement near the beginning of the program. This parameter is set at KP = 1 and specifies that the Lennard-Jones potential should be used. If any other value is set for KP then the repulsive potential is used.

2.2. Automatic switch of absolute tolerance parameter

For potentials with an attractive part the phase shift may suddenly jump by π . The correct solution of such a problem can require the use of extremely small step sizes in the solving of the differential equations. Because of this we have used the LSODA stiff equation solver method to address the problem. In order to properly describe the sharp jumps of the phase shift that are encountered one must set the absolute tolerance parameter *atol* equals to 10^{-14} in the LSODA computer code. For sufficiently high energies and orbital quantum numbers where quasi-bound energy levels no longer exist *atol* may safely be set to the higher

value of 10^{-11} . This value of *atol* may also be used for a purely repulsive interatomic potential.

The function *Ecr* determines the value of the critical energy above which the greater value of *atol* may be used.

2.3. Choice of control parameters

The program can operate in two different ways governed by the two possible control words ESCAN or LSCAN in the input data. The control word ESCAN instructs the program to perform a scan over energies for a fixed value to the orbital angular momentum quantum number *l*. While the control word LSCAN instructs the program to perform a scan over orbital angular momentum quantum numbers for a fixed value to the relative collision energy. The input file should be given the name VPA_input.txt. The main output appears in the file LPH.TXT. This lists the computed value of the phase shift for particular *l* values and energies. It also gives the partial cross sections. The sum of these over *l* yields the scattering cross section [9].

ESCAN

A typical set of input data following the ESCAN control word is:

```
ESCAN
0          !l – specifies the value of the angular
           momentum quantum number
1.5d0,60.0d0 !Emin, Emax – specifies the minimum and
           maximum collision energies of the energy
           scan, values given in cm-1. The program
           evaluates the phase shift for 120 values of
           the energy between the specified limits.
```

LSCAN

A typical set of input data following the LSCAN control word is:

```
LSCAN
0,20      !LMIN,LMAX specifies the low and high limits of
           the scan over orbital angular momentum
           quantum numbers l
10.0      ! E specifies the fixed value of the collision
           energy given in cm-1
```

2.4. Subroutine InitPoint

The key problem in implementing the phase amplitude method is the choice of an initial value for the phase shift at small internuclear or *R* values. We present a universal solution for this problem in our subroutine InitPoint. Our approach is to fix just one initial value of the phase shift, equal to 10^{-10} , and calculate initial values of $x = k \times R$ for all necessary *l*. From $x = k \times R$, *R* is then defined immediately for an arbitrary *E*. That is why it is universal and works for all *l* and all potentials. The subroutine InitPoint(Y0,Lmax,Xmax,RW) has three input parameters Y0, Lmax and Xmax and two output ones Lmax and RW. Y0 is a starting value of phase shifts for all *l*. XMAX is the Maximum allowed value of *kR*. The subroutine computes an array RW which holds the starting values of $k \times R$ for the solution of Eq. (10) for all orbital quantum numbers *l*. The subroutine may reduce the input value of Lmax so as to ensure that the maximum value of *R* is always in the classically allowed region.

2.5. Function Ecr

The DOUBLE PRECISION FUNCTION Ecr(RMU,Rmax,KP) has three input parameters RMU, Rmax and KP. RMU is the reduced mass of the interacting particles and Rmax is the maximal radii of the interatomic potential. The function finds critical values of the orbital angular momentum *l* and of the relative kinetic energy *E*. Only the critical value of the energy is passed back to the main program. Lcr is the value of the angular momentum above which the effective potential no longer has a minimum for a potential with an attractive part. The effective potential is the real potential plus the centrifugal barrier. Ecr is the critical value of the relative kinetic energy such that there are no shape resonances at higher energies. Shape resonances occur when the kinetic energy coincides with a quasi-bound state trapped inside effective potential.

2.6. Subroutine F1 - phase shift derivative

The subroutine f1(KP, l,WK,neq,R,Y,YP) computes the derivative of the phase shift as required in Eq. (10). It has six input parameters KP, *l*, WK, neq, R and Y and outputs one quantity, the derivative of the phase shift, YP. *l* is the orbital angular momentum quantum number, WK is the wave vector, R is the internuclear separation and Y is the phase shift.

2.7. Subroutine Jac1 - Jacobian of phase the shift derivative

The subroutine Jac1 calculates the Jacobian of Eq. (22). Its argument list is Jac1(KP, *l*, WK, neq, t, y, pd, nrowpd). *l* is the orbital angular momentum quantum number, WK is the wave vector, t is the internuclear separation and y is the phase shift.

The variable pd is the Jacobian calculated using Eqs. (22) and (23) and is output by the subroutine. The integer neq is the dimension of the array WK, and nrowpd is the row dimension of pd, both of these dimensions are 1 in the present application.

2.8. Subroutine WKB

At higher energies the semiclassical WKB method [9] (sometimes referred to as the JWKB method) becomes increasingly valid. At energies greater than $10 \times E_{cr}$ the main program calls the subroutine WKB(KP,Rmu,WK, l,ax,bx,Pwkb) to compute phase shifts using the WKB approximation. These are output to file LPH.TXT along with the other computed phase shifts. It has seven input parameters KP, Rmu, WK, *l*, ax, bx, and Pwkb and outputs one quantity, the WKB phase shift, Pwkb. *l* is the orbital angular momentum quantum number, WK is the wave vector, ax and bx are the initial and final internuclear separations.

3. Validation and testing of the program

We have validated our VPA program by using it to calculate the phase shifts for the atom–atom scattering of two Ne atoms and comparing the results with those obtained using the WKB approximation. This approximation works well at higher energies (see Appendix F of Ref. [9]). In the material provided with the program, we present a few phase shifts calculated for Ne–Ne scattering at a relative collision energy of $100\,000\text{ cm}^{-1}$ (12.4 eV) using two potentials, one with and one without a well and computed using both our VPA method and the WKB approximation. It can be seen that WKB method gives absolute values of phase shifts in reasonably good agreement with our VPA results.

In addition, to further validate our VPA code, we have performed calculations of electron–atom scattering phase shifts to compare with the results of Cash et al. [10] based on the numerical solution of the Schrödinger equation followed by an

analysis of the asymptotic form of the scattering wavefunction. The authors of Ref. [10] obtained phase shifts of -0.48308134 and 1.41480963 for a wave vector $k = 5$ and orbital quantum numbers $l = 0$ and 10 . Calculations using our VPA program yield values of -0.4830254 and 4.556453 for these phase shifts. By subtracting π from the last value one gets 1.41486 . As we can see the difference between our results and those given in Ref. [10] is in the 5th significant figure, which seems entirely satisfactory.

We have also tested the program by reproducing some of the diagrams of phase shifts in our book on collision theory [9]. Using the following input data to the program:

```
ESCAN
12 !L
4.9d0,5.5d0 !Emin, Emax in CM-1
```

We obtain the following output in the file LPH.txt (only the first few lines are shown):

```
Kinetic Energy/cm-1, L, Phase shift/pi, Partial Cross section
E(1/cm) = 4.89926 L = 12 Phase = 0.1151604E+00 PCS = 0.1355E+02
E(1/cm) = 4.90426 L = 12 Phase = 0.1173496E+00 PCS = 0.1403E+02
E(1/cm) = 4.90926 L = 12 Phase = 0.1199212E+00 PCS = 0.1460E+02
E(1/cm) = 4.91426 L = 12 Phase = 0.1229928E+00 PCS = 0.1531E+02
E(1/cm) = 4.91926 L = 12 Phase = 0.1267508E+00 PCS = 0.1619E+02
E(1/cm) = 4.92426 L = 12 Phase = 0.1314707E+00 PCS = 0.1733E+02
E(1/cm) = 4.92926 L = 12 Phase = 0.1376015E+00 PCS = 0.1886E+02
E(1/cm) = 4.93426 L = 12 Phase = 0.1459280E+00 PCS = 0.2103E+02
E(1/cm) = 4.93926 L = 12 Phase = 0.1579281E+00 PCS = 0.2430E+02
E(1/cm) = 4.94426 L = 12 Phase = 0.1767584E+00 PCS = 0.2978E+02
E(1/cm) = 4.94926 L = 12 Phase = 0.2104177E+00 PCS = 0.4034E+02
E(1/cm) = 4.95425 L = 12 Phase = 0.2849135E+00 PCS = 0.6510E+02
E(1/cm) = 4.95925 L = 12 Phase = 0.4989576E+00 PCS = 0.1068E+03
E(1/cm) = 4.96425 L = 12 Phase = 0.8198521E+00 PCS = 0.3068E+02
E(1/cm) = 4.96925 L = 12 Phase = 0.9491084E+00 PCS = 0.2702E+01
E(1/cm) = 4.97425 L = 12 Phase = 0.9990245E+00 PCS = 0.1000E-02
E(1/cm) = 4.97925 L = 12 Phase = 0.1024260E+01 PCS = 0.6168E+00
E(1/cm) = 4.98425 L = 12 Phase = 0.1039366E+01 PCS = 0.1617E+01
E(1/cm) = 4.98925 L = 12 Phase = 0.1049421E+01 PCS = 0.2539E+01
E(1/cm) = 4.99425 L = 12 Phase = 0.1056612E+01 PCS = 0.3320E+01
E(1/cm) = 4.99925 L = 12 Phase = 0.1062026E+01 PCS = 0.3973E+01
E(1/cm) = 5.00425 L = 12 Phase = 0.1066265E+01 PCS = 0.4522E+01
E(1/cm) = 5.00925 L = 12 Phase = 0.1069687E+01 PCS = 0.4988E+01
E(1/cm) = 5.01425 L = 12 Phase = 0.1072517E+01 PCS = 0.5389E+01
E(1/cm) = 5.01924 L = 12 Phase = 0.1074907E+01 PCS = 0.5738E+01
E(1/cm) = 5.02424 L = 12 Phase = 0.1076958E+01 PCS = 0.6044E+01
E(1/cm) = 5.02924 L = 12 Phase = 0.1078746E+01 PCS = 0.6316E+01
E(1/cm) = 5.03424 L = 12 Phase = 0.1080322E+01 PCS = 0.6559E+01
E(1/cm) = 5.03924 L = 12 Phase = 0.1081728E+01 PCS = 0.6779E+01
E(1/cm) = 5.04424 L = 12 Phase = 0.1082994E+01 PCS = 0.6979E+01
E(1/cm) = 5.04924 L = 12 Phase = 0.1084144E+01 PCS = 0.7162E+01
E(1/cm) = 5.05424 L = 12 Phase = 0.1085196E+01 PCS = 0.7331E+01
E(1/cm) = 5.05924 L = 12 Phase = 0.1086165E+01 PCS = 0.7487E+01
E(1/cm) = 5.06424 L = 12 Phase = 0.1087063E+01 PCS = 0.7632E+01
E(1/cm) = 5.06924 L = 12 Phase = 0.1087900E+01 PCS = 0.7768E+01
E(1/cm) = 5.07424 L = 12 Phase = 0.1088684E+01 PCS = 0.7896E+01
E(1/cm) = 5.07924 L = 12 Phase = 0.1089422E+01 PCS = 0.8017E+01
E(1/cm) = 5.08423 L = 12 Phase = 0.1090119E+01 PCS = 0.8131E+01
E(1/cm) = 5.08923 L = 12 Phase = 0.1090780E+01 PCS = 0.8239E+01
```

The first column contains the collision energy, the second the phase shift divided by π and the third the partial cross section. In Fig. 1 these results are plotted.

The figure shows the sharp increase by π of the phase shift for Ne–Ne scattering on an effective potential corresponding to an angular momentum quantum number of $l = 12$, at around a relative collision energy of 4.95 cm^{-1} . These calculations took immense trouble and many hours of computer time to perform at the time of the writing of the book. The current program produces these results in less than two seconds on a three year old desktop computer. Other illustrative examples from our book have also been reproduced.

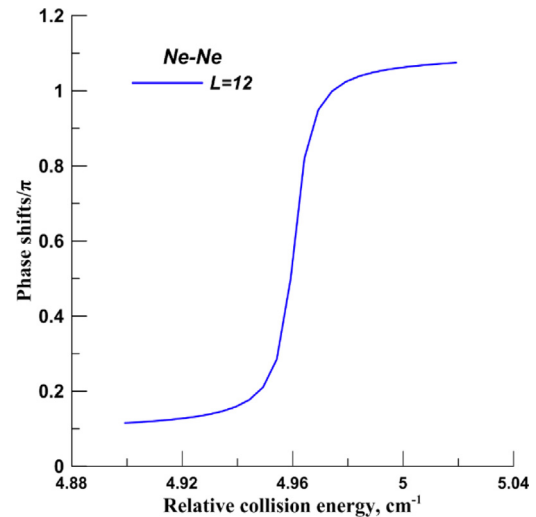


Fig. 1. Plot of the phase shift for a collision of two Ne atoms in the vicinity of a shape resonance.

4. Limitations of the program

The VPA code can cope with all possible values of the orbital angular momentum quantum number l . As l increases to high values the phase shift decreases towards zero. In the mode of operation when the program is scanning over the orbital angular momentum quantum numbers (control word LSCAN) the scan will stop when the phase shift decreases below a value YEND which is specified near the start of the main program. This parameter is presently set at $YEND = 0.0001$ but can be changed by a user. The maximal value of l is defined in subroutine InitPoint to be 10000 however in can be increased if needed.

However, there are some limits on the collision energies. Currently, the program is tested for energies from 1.5 cm^{-1} to $700,000 \text{ cm}^{-1}$. One of the reasons for the upper limit on the energy is the potential routine. This routine has a cut-off for internuclear separations less than 2.5 a.u. The potential energy at this small internuclear separation is very large, lying high up on the repulsive wall of the potential. To use higher collision energies the potential must also extend to comparable energies, i.e. smaller internuclear separations. At energies below 1.5 cm^{-1} the code gives a non-smooth dependence of phase shifts versus energy mainly for $l = 1-10$. However, for $l = 0$ there is no problem at even extremely low energies up to 10^{-5} cm^{-1} .

Fig. 2 demonstrates dependence of the phase shift values on energies between 10^{-5} and 0.1 cm^{-1} . One sees that the phase shift reaches a value of 3π as it should as the energy approaches zero. This is because the Lennard-Jones potential used supports 3 bound states. We are currently attempting to improve the code so that it produces reliable results in the $0-1.5 \text{ cm}^{-1}$ energy range for all l values.

5. Conclusion

The program represents an efficient, viable implementation of the Variable Phase Approach (VPA) to the calculation of elastic scattering phase shifts. There exists no other method for the direct calculation of the phase shift and there are no other generally distributed computer codes for the calculation of scattering phase shifts. The computer code works for both singular and non-singular potentials over a wide range of relative kinetic energies. The main technical feature of the program is the secure calculation of the initial point for the numerical integration which leads to the program generating a smooth dependence of the

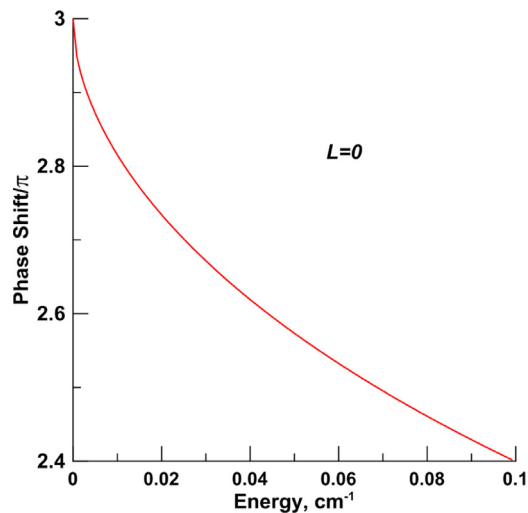


Fig. 2. Dependence of the phase shift values on energies for $l = 0$ for attractive potential.

phase shifts on l and E . In addition, our fast and robust algorithm for the calculation of the right hand side of the variable phase equation (Eq. (10)) and the automatic choice of the absolute tolerance parameter based on the analysis of the interaction potential increases the efficiency of the VPA program.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cpc.2021.107895>.

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